Solution: Sliding Window Median

Problem Statement

Given an array of numbers and a number ‘k’, find the median of all the ‘k’ sized sub-arrays (or windows) of the array.

Example 1:

Input: nums=[1, 2, -1, 3, 5], k = 2
Output: [1.5, 0.5, 1.0, 4.0]
Explanation: Lets consider all windows of size ‘2’:

[**1, 2**, -1, 3, 5] -> median is 1.5
[1, **2, -1**, 3, 5] -> median is 0.5
[1, 2, **-1, 3**, 5] -> median is 1.0
[1, 2, -1, **3, 5**] -> median is 4.0

Example 2:

Input: nums=[1, 2, -1, 3, 5], k = 3
Output: [1.0, 2.0, 3.0]
Explanation: Let's consider all windows of size ‘3’:

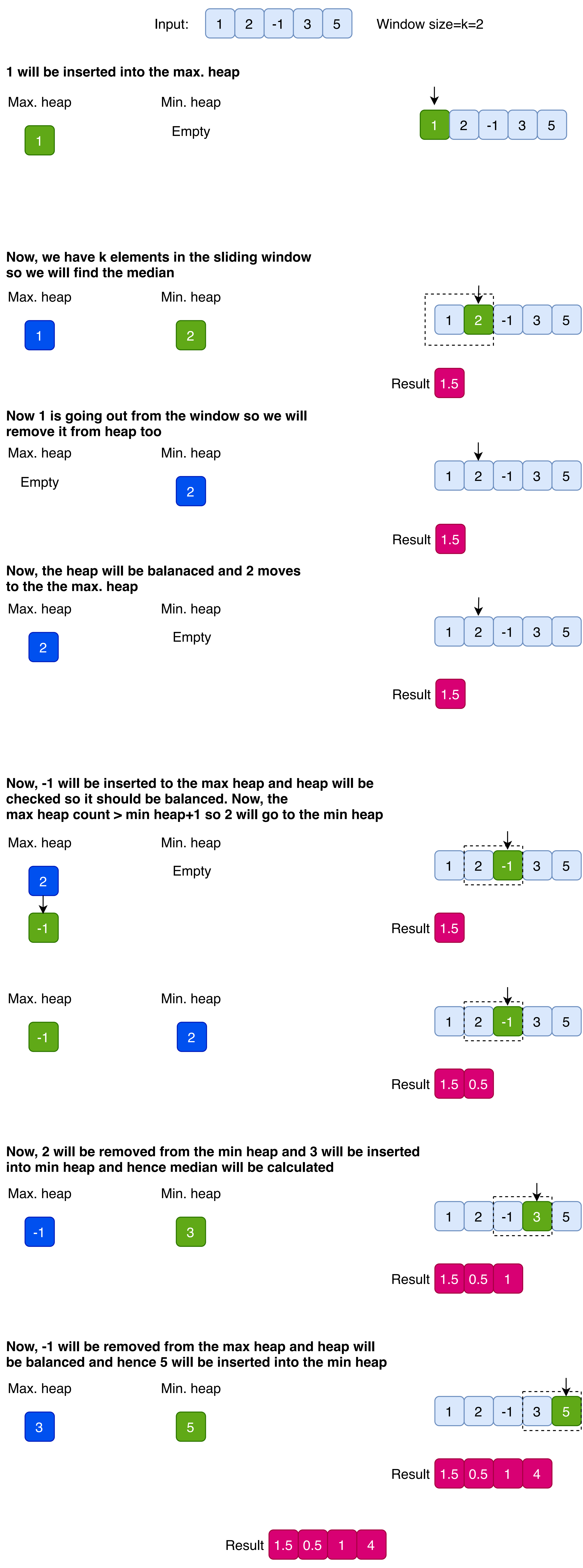
[**1, 2, -1**, 3, 5] -> median is 1.0
[1, **2, -1, 3**, 5] -> median is 2.0
[1, 2, **-1, 3, 5**] -> median is 3.0

Solution

This problem follows the **Two Heaps** pattern and share similarities with **Find the Median of a Number Stream.** We can follow a similar approach of maintaining a max-heap and a min-heap for the list of numbers to find their median.

The only difference is that we need to keep track of a sliding window of ‘k’ numbers. This means, in each iteration, when we insert a new number in the heaps, we need to remove one number from the heaps which is going out of the sliding window. After the removal, we need to rebalance the heaps in the same way that we did while inserting.

Here is the visual representation of the algorithm:



1 will be inserted into the max. heap

Now, we have k elements in the sliding window so we will find the median

Now 1 is going out from the window so we will remove it from heap too

Now, the heap will be balanaced and 2 moves to the the max. heap

Now, -1 will be inserted to the max heap and heap will be checked so it should be balanced. Now, the max heap count > min heap+1 so 2 will go to the min heap

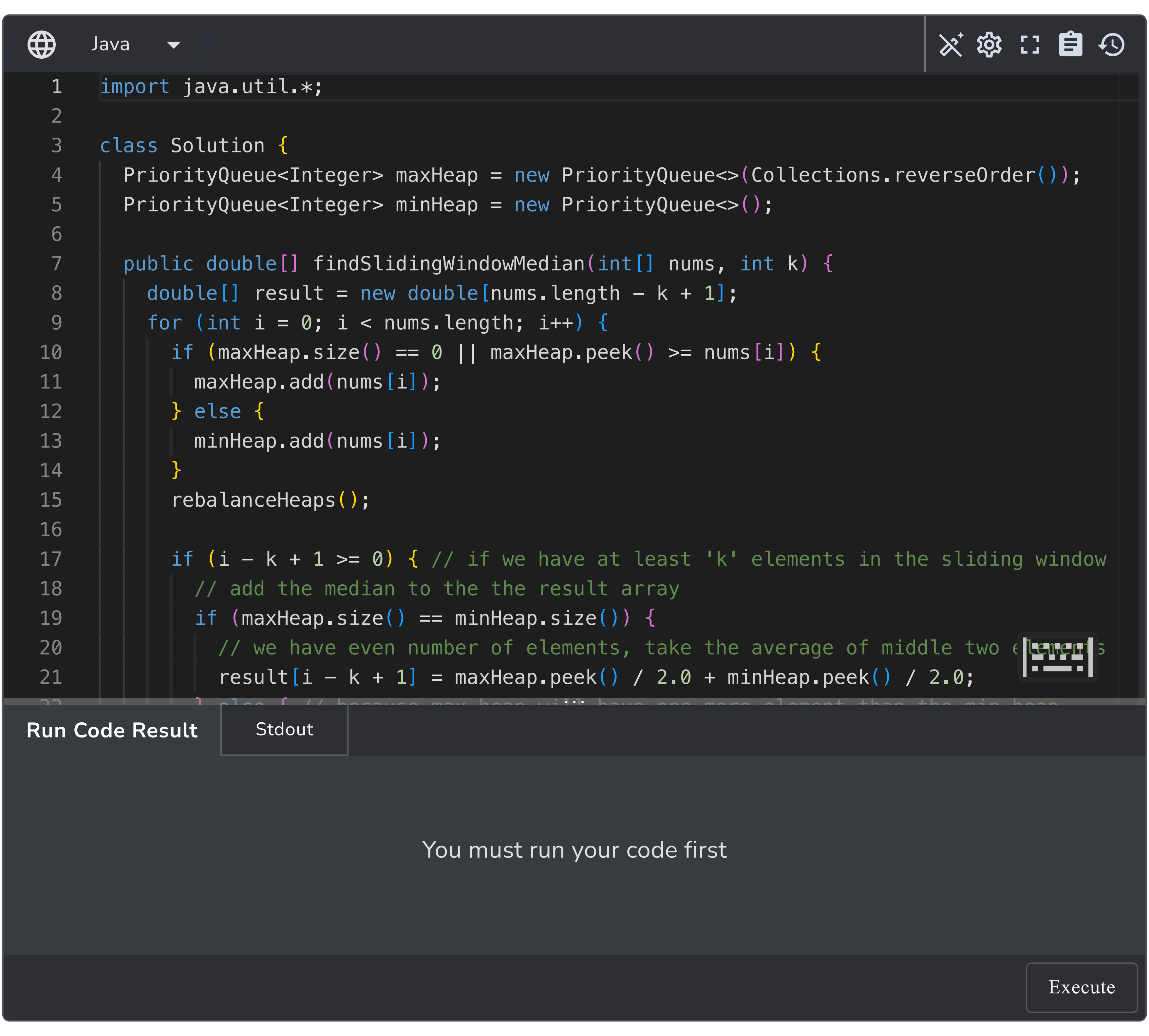
Now, 2 will be removed from the min heap and 3 will be inserted into min heap and hence median will be calculated

Now, -1 will be removed from the max heap and heap will be balanced and hence 5 will be inserted into the min heap

Sliding Window Median

Code

Here is the code for this algorithm:



```java
import java.util.*;

class Solution {
  PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
  PriorityQueue<Integer> minHeap = new PriorityQueue<>();

  public double[] findSlidingWindowMedian(int[] nums, int k) {
    double[] result = new double[nums.length - k + 1];
    for (int i = 0; i < nums.length; i++) {
      if (maxHeap.size() == 0 || maxHeap.peek() >= nums[i]) {
        maxHeap.add(nums[i]);
      } else {
        minHeap.add(nums[i]);
      }
      rebalanceHeaps();

      if (i - k + 1 >= 0) { // if we have at least 'k' elements in the sliding window
        // add the median to the the result array
        if (maxHeap.size() == minHeap.size()) {
          // we have even number of elements, take the average of middle two elements
          result[i - k + 1] = maxHeap.peek() / 2.0 + minHeap.peek() / 2.0;
```

You must run your code first

Time Complexity

The time complexity of our algorithm is $O(N * K)$ where ‘N’ is the total number of elements in the input array and ‘K’ is the size of the sliding window. This is due to the fact that we are going through all the ‘N’ numbers and, while doing so, we are doing two things:

1. Inserting/removing numbers from heaps of size ‘K’. This will take $O(logK)$.
2. Removing the element going out of the sliding window. This will take $O(K)$ as we will be searching this element in an array of size ‘K’ (i.e., a heap).

Space Complexity

Ignoring the space needed for the output array, the space complexity will be $O(K)$ because, at any time, we will be storing all the numbers within the sliding window.

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